Exercise

1. Number Checker:
   - Write a script named num_check.sh that takes one number as an argument.
   - The script should use an if...elif...else statement to print whether the number is positive, negative, or zero.

```
hp@TF-COMP-786:~/my_project$ cat num_check.sh
#!/bin/bash

# Check if user gave an argument
if [ -z "$1" ]; then
  echo "Usage: $0 <number>"
  exit 1
fi

NUM=$1
if [ "$NUM" -gt 0 ]; then
  echo "The number is positive."
elif [ "$NUM" -lt 0 ]; then
  echo "The number is negative."
else
  echo "The number is zero."
fi
hp@TF-COMP-786:~/my_project$ ./num_check.sh -3
The number is negative.
```

2. File Inspector:
   - Write a script named file_info.sh that takes one argument (a name of something in the current directory).
   - The script should use if and file operators (-f, -d) to check and print whether the argument is a regular file, a directory, or does not exist.

```
hp@TF-COMP-786:~/my_project$ cat file_info.sh
#!/bin/bash

if [ -z "$1" ]; then
  echo "Usage: $0 <filename or directory>"
  exit 1
fi

TARGET=$1
if [ -f "$TARGET" ]; then
  echo "'$TARGET' is a regular file."
elif [ -d "$TARGET" ]; then
  echo "'$TARGET' is a directory."
else
  echo "'$TARGET' does not exist."
fi
hp@TF-COMP-786:~/my_project$ ./file_info.sh
'.' is a directory.
```

3. Simple Animal Facts:
   - Write a script named facts.sh that accepts one argument: "cat", "dog", or "bird".
   - Use a case statement to print a simple fact for the given animal. Include a default case for any other input.

```
hp@TF-COMP-786:~$ cat > facts.sh
#!/bin/bash

case "$1" in
  cat)
    echo "Cats sleep up to 16 hours a day!"
    ;;
  dog)
    echo "Dogs can hear sounds up to four times farther than humans."
    ;;
  bird)
    echo "Some birds migrate thousands of miles every year."
    ;;
  *)
    echo "I don't have a fact about that animal."
    ;;
esac
hp@TF-COMP-786:~$ chmod +x facts.sh
hp@TF-COMP-786:~$ ./facts.sh dog
Dogs can hear sounds up to four times farther than humans.
```

4. Countdown Timer:
    - Write a script named countdown.sh that uses a C-style for loop to count down from 10 to 1, printing each number. After the loop, it should print "Liftoff!".

```
hp@TF-COMP-786:~$ cat > countdown.sh
#!/bin/bash

for (( i=10; i>=1; i-- ))
do
  echo "$i"
  sleep 1
done

echo "Liftoff!"
hp@TF-COMP-786:~$ chmod +x countdown.sh
```

```
hp@TF-COMP-786:~$ ./countdown.sh
10
9
8
7
6
5
4
3
2
1
Liftoff!
```

5. Summation Calculator:
    - Write a script named sum.sh that takes one number as an argument.
    - Using a while loop and the $((...)) syntax for arithmetic, calculate and print the sum of all integers from 1 up to that number. (e.g., if the input is 5, the output should be 15, which is 1+2+3+4+5).

```
hp@TF-COMP-786:~$ cat > sum.sh
#!/bin/bash

if [ -z "$1" ]; then
  echo "Usage: $0 <number>"
  exit 1
fi

NUM=$1
SUM=0
i=1

while [ $i -le $NUM ]
do
  SUM=$((SUM + i))
  i=$((i + 1))
done

echo "The sum of numbers from 1 to $NUM is $SUM
hp@TF-COMP-786:~$ chmod +x sum.sh
hp@TF-COMP-786:~$ ./sum.sh 5
The sum of numbers from 1 to 5 is 15.
```

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6. Greeting Function:
    - Write a script named greeter.sh that defines a function called say_hello.
    - The function should take one argument (a name) and print "Welcome to the team,[name]!".
    - Call the function three times from within the script, with three different names.

```
hp@TF-COMP-786:~$ cat > greeter.sh
#!/bin/bash

say_hello() {
  echo "Welcome to the team, $1!"
}

say_hello "Tooba"
say_hello "Ali"
say_hello "Sara"
```

```
hp@TF-COMP-786:~$ chmod +x greeter.sh
hp@TF-COMP-786:~$ ./greeter.sh
Welcome to the team, Tooba!
Welcome to the team, Ali!
Welcome to the team, Sara!
```